Advanced Business Analytics Modelling

Airbnb Oregon Salem Data Analysis

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Airbnb Salem Oregon Data Analysis

Objective

The primary objective of this project is to develop and evaluate predictive models for Airbnb Price using statistical and machine learning techniques. The project is designed to explore the relationships between various predictors and the target variable, and to evaluate the performance of different modeling approaches in terms of accuracy and interpretability. This project involves a structured approach to data exploration, preprocessing, model building, and performance evaluation. The dataset was obtained from the http://insideairbnb.com/explore/.

Data Exploration

This section describes the preliminary investigation and preprocessing procedures carried out on the Salem, Oregon, Airbnb listings dataset. Before starting the modeling process, it was important to understand the distribution, quality, and structure of the data. The data contains 79 variables and 351 observations.

The CONTENTS Procedure

Data Set Name	WORK.LISTINGS	Observations	351
Member Type	DATA	Variables	79
Engine	V9	Indexes	0
Created	10/30/2025 17:56:16	Observation Length	6216
Last Modified	10/30/2025 17:56:16	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Fig 1: Data Sets Exploration

Geospatial Distribution

To understand and visualize the spread of Airbnb listings across Salem, Oregon geographical maps were created using PROC SGMAP. The map revealed spatial clustering and helped define urban, suburban, and residential zones for further analysis. There are a total of 351 Airbnb listings, and the map shows that Ward 1 and Ward 2 have the greatest number of listings among all eight wards.

Fig 1: Oregon, Salem Source: Wikipedia

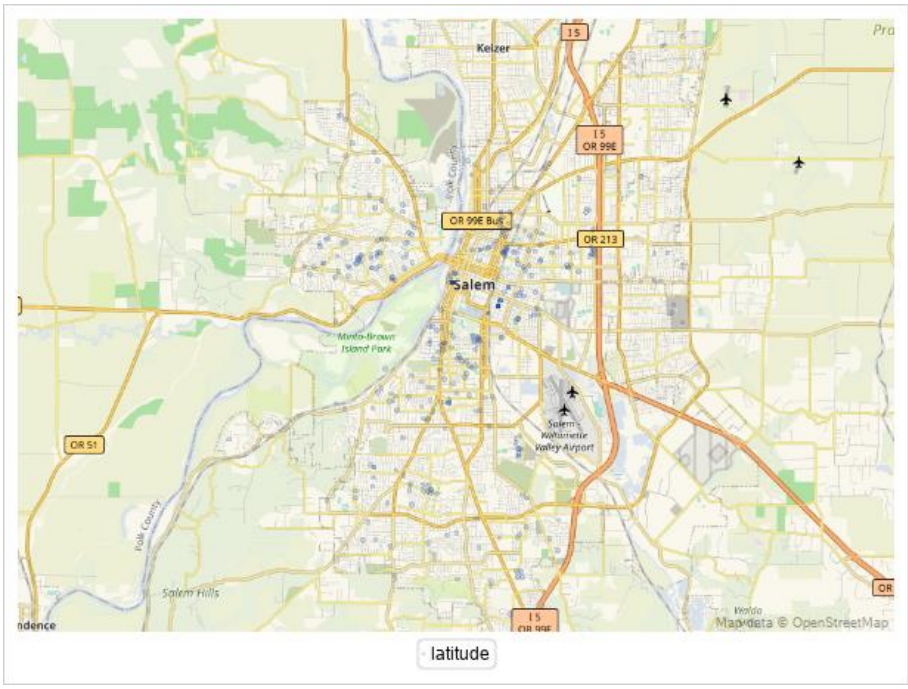

Fig 2: Geographic Distribution of Airbnb Listings in Salem, Oregon

Fig 3: Airbnb listings across wards in Salem, Oregon

Target Variable

The target variable for the project is price, which represents the cost of Airbnb listing in Salem, Oregon, USA. To investigate the dependent variable distribution histogram was plotted to see the raw distribution which showed the significant right skewness. Similarly, to address this skewness, a log transformation was applied, resulting in a more normalized distribution. This transformation was performed using PROC UNIVARIATE and histogram plots.

Price Distribution

The UNIVARIATE Procedure
Variable: price

Moments			
N	276	Sum Weights	276
Mean	123.855072	Sum Observations	34184
Std Deviation	71.8322845	Variance	5159.8771
Skewness	1.68578507	Kurtosis	4.04170662
Uncorrected SS	5652828	Corrected SS	1418966.2
Coeff Variation	57.997047	Std Error Mean	4.32379543

Basic Statistical Measures			
Location		Variability	
Mean	123.8551	Std Deviation	71.83228
Median	108.5000	Variance	5160
Mode	99.0000	Range	445.00000
		Interquartile Range	77.00000

Fig 4: Basic statistical measures of target variable 'price'

The PROC UNIVARIATE output shows that there is a significant difference between the mean and median values which hints us about the presence of outliers which affects the mean value. Likewise, the range gives us a difference between the highest and lowest value, which is 445 range per price.

Quantiles (Definition 5)	
Level	Quantile
100% Max	475.0
99%	378.0
95%	270.0
90%	221.0
75% Q3	152.0
50% Median	108.5
25% Q1	75.0
10%	49.0
5%	43.0
1%	30.0
0% Min	30.0

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
30	39	350	9
30	29	362	108
30	27	378	80
33	221	452	207
34	266	475	81

Fig 5: PROC UNIVARIATE Quartiles, extreme observations

The above figure shows the Quantiles of the variables, which demonstrates the minimum and maximum values at certain percentage cutoff points. From the table, it was found that the minimum value is $0 and the maximum value for the price is $475.

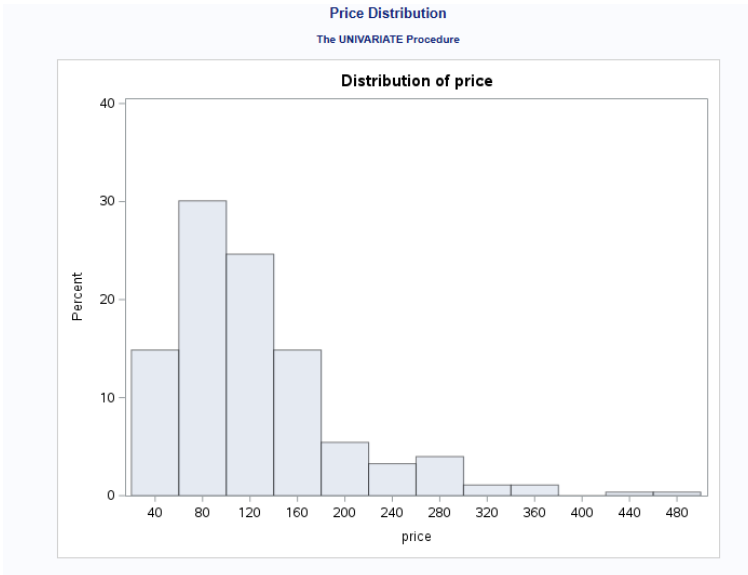

Fig 6: Histogram of the dependent variable price

The histogram shows a highly skewed distribution with a right-sided tail. Most listings fall within the $40–$160 range, while a small number of listings show significantly higher prices, creating a long tail. This skewness suggests the need for a log transformation to normalize the target variable for modeling purposes.

Log adjusted target value

To address the issue of positive skewness, Logarithmic transformation was applied to the Price variable, creating a new variable called log_price. To reduce this skewness, extreme outliers were removed by filtering listings priced below $30 and above $475. This transformation produced a normal distribution, making it suitable for linear regression and LASSO.

Log-Transformed Price Distribution

The UNIVARIATE Procedure
Variable: log_price

Moments			
N	276	Sum Weights	276
Mean	4.6719334	Sum Observations	1289.45362
Std Deviation	0.54429209	Variance	0.29625388
Skewness	0.0017085	Kurtosis	-0.133111
Uncorrected SS	6105.71125	Corrected SS	81.4698163
Coeff Variation	11.6502536	Std Error Mean	0.03276253

Basic Statistical Measures			
Location		Variability	
Mean	4.671933	Std Deviation	0.54429
Median	4.686740	Variance	0.29625
Mode	4.595120	Range	2.76212
		Interquartile Range	0.70639

Fig 7: PROC UNIVARIATE moments table on Log adjusted target variable

Quantiles (Definition 5)	
Level	Quantile
100% Max	475.0
99%	378.0
95%	270.0
90%	221.0
75% Q3	152.0
50% Median	108.5
25% Q1	75.0
10%	49.0
5%	43.0
1%	30.0
0% Min	30.0

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
30	39	350	9
30	29	362	108
30	27	378	80
33	221	452	207
34	266	475	81

Fig 8: Quantile table on Log adjusted target variable

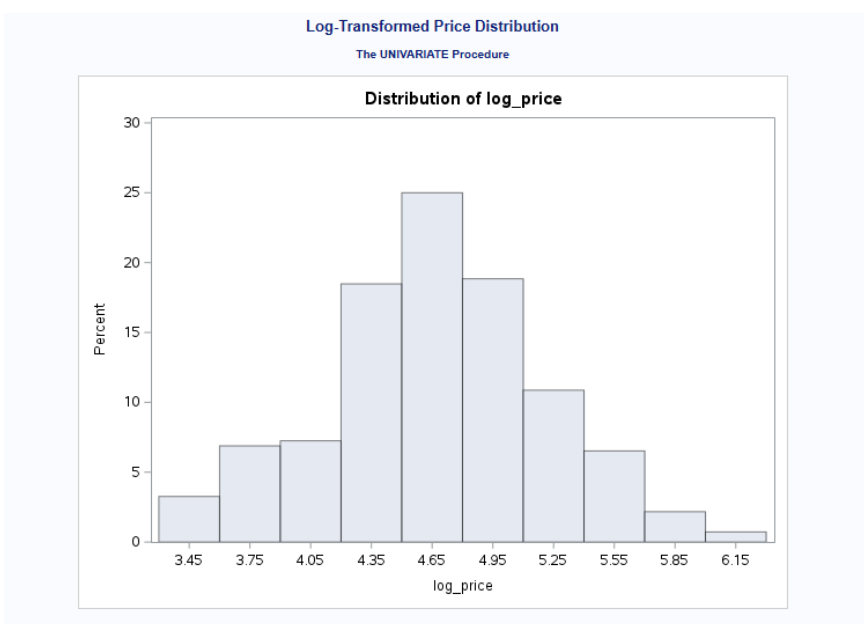

Fig 9: Histogram of logarithmic transformed log_price

To address the extreme outliers and nonsensical values, the dependent variable was transformed into a normally distributed variable as depicted in the histogram.

Numerical variable analysis

To analyze the numerical variables, PROC MEANS was used to compute key descriptive statistics, including count, number of missing values, minimum, maximum, mean, median, and standard deviation. This analysis provided insights into data quality, highlighted variables with missing values, and helped identify potential outliers for further investigation.

Variable	Label	N	N Miss	Minimum	Maximum	Mean	Median	Std Dev
id	id	276	0	199568.00	1.5125209E18	7.5660465E17	8.5947197E17	5.4501861E17
scrape_id	scrape_id	276	0	2.0250925E13	2.0250925E13	2.0250925E13	2.0250925E13	0
last_scraped	last_scraped	276	0	24009.00	24009.00	24009.00	24009.00	0
host_id	host_id	276	0	183514.00	707469513	232798631	161820068	200315605
host_since	host_since	276	0	18471.00	23937.00	21423.67	21161.00	1200.40
host_listings_count	host_listings_count	276	0	1.0000000	1231.00	19.4891304	2.0000000	97.7886583
host_total_listings_count	host_total_listings_count	276	0	1.0000000	2787.00	29.2898551	3.0000000	193.9583349
latitude	latitude	276	0	44.8586400	44.9998965	44.9308252	44.9381880	0.0265626
longitude	longitude	276	0	-123.1181337	-122.9613985	-123.0369486	-123.0331247	0.0279135
accommodates	accommodates	276	0	1.0000000	16.0000000	4.0398551	4.0000000	2.5659034
bathrooms	bathrooms	276	0	0	4.0000000	1.4148551	1.0000000	0.6815733
bedrooms	bedrooms	276	0	0	6.0000000	1.8623188	1.0000000	1.1131742
beds	beds	276	0	0	10.0000000	2.1992754	2.0000000	1.4991390
minimum_nights	minimum_nights	276	0	1.0000000	60.0000000	11.7898551	2.0000000	14.8573596
maximum_nights	maximum_nights	276	0	2.0000000	1125.00	414.4021739	365.0000000	361.6565440
minimum_minimum_nights	minimum_minimum_nights	276	0	1.0000000	60.0000000	10.9311594	2.0000000	14.4915890
maximum_minimum_nights	maximum_minimum_nights	276	0	1.0000000	60.0000000	13.0543478	3.0000000	15.0644898
minimum_maximum_nights	minimum_maximum_nights	276	0	1.0000000	1125.00	602.0797101	365.0000000	438.9120858
maximum_maximum_nights	maximum_maximum_nights	276	0	2.0000000	1125.00	619.3333333	365.0000000	425.3705949
minimum_nights_avg_ntm	minimum_nights_avg_ntm	276	0	1.0000000	60.0000000	12.5778986	2.4500000	14.8140419
maximum_nights_avg_ntm	maximum_nights_avg_ntm	276	0	2.0000000	1125.00	608.5184783	365.0000000	434.5643638
availability_30	availability_30	276	0	0	30.0000000	13.8659420	14.0000000	11.0427335
availability_60	availability_60	276	0	0	60.0000000	34.4855072	38.0000000	20.9330484
availability_90	availability_90	276	0	0	90.0000000	55.7862319	64.0000000	30.4706349
availability_365	availability_365	276	0	0	365.0000000	229.3260870	259.0000000	110.9567426
calendar_last_scraped	calendar_last_scraped	276	0	24009.00	24009.00	24009.00	24009.00	0
number_of_reviews	number_of_reviews	276	0	0	1019.00	70.5144928	22.5000000	135.6136349
number_of_reviews_ltm	number_of_reviews_ltm	276	0	0	143.0000000	17.1231884	7.5000000	23.1686160
number_of_reviews_l30d	number_of_reviews_l30d	276	0	0	15.0000000	1.4094203	0	2.3329540
availability_eoy	availability_eoy	276	0	0	98.0000000	60.8913043	69.0000000	32.5658348
number_of_reviews_ly	number_of_reviews_ly	276	0	0	134.0000000	15.6992754	4.5000000	23.7528211
estimated_occupancy_l365d	estimated_occupancy_l365d	276	0	0	255.0000000	117.8985507	102.0000000	101.9590855
estimated_revenue_l365d	estimated_revenue_l365d	276	0	0	70560.00	14200.74	10828.50	14450.10
first_review	first_review	235	41	18868.00	24006.00	22878.99	23083.00	915.7107906
last_review	last_review	235	41	21052.00	24008.00	23885.14	23983.00	365.2493884
review_scores_rating	review_scores_rating	235	41	2.0000000	5.0000000	4.8669362	4.9400000	0.2779688
review_scores_accuracy	review_scores_accuracy	235	41	3.0000000	5.0000000	4.8775745	4.9500000	0.2416814
review_scores_cleanliness	review_scores_cleanliness	235	41	3.0000000	5.0000000	4.8612340	4.9500000	0.2450758
review_scores_checkin	review_scores_checkin	235	41	3.0000000	5.0000000	4.9265532	4.9700000	0.1826667
review_scores_communication	review_scores_communication	235	41	2.0000000	5.0000000	4.9189787	4.9700000	0.2392827
review_scores_location	review_scores_location	235	41	2.5000000	5.0000000	4.8021702	4.8800000	0.3107792
review_scores_value	review_scores_value	235	41	1.0000000	5.0000000	4.8004681	4.8800000	0.3407235
calculated_host_listings_count	calculated_host_listings_count	276	0	1.0000000	15.0000000	3.4601449	2.0000000	3.8264921
calculated_host_listings_count_e	calculated_host_listings_count_entire_homes	276	0	0	14.0000000	2.6086957	1.0000000	3.6470357
calculated_host_listings_count_p	calculated_host_listings_count_private_rooms	276	0	0	7.0000000	0.6666667	0	1.5007069
calculated_host_listings_count_s	calculated_host_listings_count_shared_rooms	276	0	0	0	0	0	0
reviews_per_month	reviews_per_month	235	41	0.0100000	10.8400000	2.0943404	1.3900000	2.0996283
price		276	0	30.0000000	475.0000000	123.8550725	108.5000000	71.8322845
log_price		276	0	3.4011974	6.1633148	4.6719334	4.6867396	0.5442921
poly_accom		276	0	1.0000000	256.0000000	22.8804348	16.0000000	30.7480704
poly_min		276	0	1.0000000	3600.00	358.9420290	4.0000000	658.0099228
log_host_listings		276	0	0.6931472	7.9330798	1.7458466	1.3862944	1.2176269
beds_per_accom		276	0	0	6.0000000	1.7605159	2.0000000	0.7003795
bath_per_accom		276	0	0	8.0000000	2.8601881	2.5000000	1.3882583

Fig: 10 Numerical Variable analysis using PROC Means

The results obtained from PROC MEANS showed critical insights into the data sets. The price ranges from a minimum of $30 to a maximum of $475 with a mean of 125.04 and a median of $139. Similarly, the standard deviation of $70.67 indicates significant variability among the price listings. The transformed variable (log_price) now spans 3.40 to 6.16, with a standard deviation of 0.51, making it more suitable for predictive modeling.

Likewise, it also illustrates the missing values in certain variables like first_review, last_review, review_scores_rating, review_scores_accuracy, review_scores_cleanliness, review_scores_checkin, review_scores_communication, review_scores_location, review_scores_value, reviews_per_month has 41 missing values comprising 15% of the dataset. However, since the numeric variables will not be available at the point of application as such we will exclude these varaibles like first_review, last_review, review_scores_rating, review_scores_accuracy, review_scores_cleanliness, review_scores_checkin, review_scores_communication, review_scores_location, review_scores_value. Likewise, we will also also eliminate the calculated_host_listings_count, calculated_host_listings_count_e, calculated_host_listings_count_p, calculated_host_listings_count_s, as they are system generated and will not be available at the point of application. Similarly, we will eliminate id, host_id and scrape_id from the models as they are unique identifiers. We will also remove latitude and longitude

The host_total_listings_count ranges from 1 to 1,231, revealing that some hosts manage thousands of properties. This large variation can influence the models so we will perform the data transformations.

Categorical variable analysis

Categorical variables were explored using PROC FREQ. Variables such as neighbourhood_cleansed, room_type, property_type, host_is_superhost, and instant_bookable were examined for frequency and missing values. This step was important for identifying high-cardinality variables and planning for level collapsing.

Missing Values - Categorical Variables

The FREQ Procedure

neighbourhood_cleansed				
neighbourhood_cleansed	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Ward 1	69	25.00	69	25.00
Ward 2	58	21.01	127	46.01
Ward 3	15	5.43	142	51.45
Ward 4	20	7.25	162	58.70
Ward 5	13	4.71	175	63.41
Ward 6	16	5.80	191	69.20
Ward 7	40	14.49	231	83.70
Ward 8	45	16.30	276	100.00

room_type				
room_type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Entire home/apt	214	77.54	214	77.54
Hotel room	9	3.26	223	80.80
Private room	53	19.20	276	100.00

Fig 11: Categorical variables neighbourhood_cleansed, room_type analysis

property_type				
property_type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Camper/RV	4	1.45	4	1.45
Entire bungalow	1	0.36	5	1.81
Entire condo	7	2.54	12	4.35
Entire cottage	3	1.09	15	5.43
Entire guest suite	18	6.52	33	11.96
Entire guesthouse	18	6.52	51	18.48
Entire home	120	43.48	171	61.96
Entire loft	4	1.45	175	63.41
Entire place	1	0.36	176	63.77
Entire rental unit	28	10.14	204	73.91
Entire townhouse	6	2.17	210	76.09
Entire vacation home	1	0.36	211	76.45
Private room in home	51	18.48	262	94.93
Private room in rental unit	1	0.36	263	95.29
Private room in townhouse	1	0.36	264	95.65
Room in hotel	9	3.26	273	98.91
Tiny home	3	1.09	276	100.00

Fig 12: Categorical variables property_type analysis

host_is_superhost				
host_is_superhost	Frequency	Percent	Cumulative Frequency	Cumulative Percent
f	105	38.04	105	38.04
t	171	61.96	276	100.00

instant_bookable				
instant_bookable	Frequency	Percent	Cumulative Frequency	Cumulative Percent
f	207	75.00	207	75.00
t	69	25.00	276	100.00

has_availability				
has_availability	Frequency	Percent	Cumulative Frequency	Cumulative Percent
t	271	100.00	271	100.00
Frequency Missing = 5				

Fig 13: Categorical variables host_is super_host, instant_bookable and has_availability analysis

The above tables illustrate the level of the categorical variables along with frequency of observations, the percentage of total observations, cumulative frequency, and cumulative percent of observations for the categorical variables. Since there are few categorical variables like

neighbourhood_cleansed, property_type which has more than 5 categories which can be problematic and were consolidated into meaningful using feature engineering.

Feature Engineering

To address high cardinality and improve model interpretability, several categorical variables were consolidated into meaningful groups. For example, the property_type variable, which originally contained numerous distinct values, was grouped into four broader categories: Entire_Place, Private_Room, Hotel_Commercial, and Specialty_Stay. Similarly, neighbourhood_cleansed was simplified into three categories based on ward characteristics: Urban (Wards 1 and 2), Residential (Wards 7 and 8), and Sub-Urban (Wards 3–6).

Continuous variables such as bedrooms, bathrooms, and beds were binned into ranges like 0–1 Bedroom, 2–3 Bedrooms, and 4–5 Bedrooms. These transformations reduce dimensionality, ensure balanced category sizes, and minimize the risk of overfitting, while preserving the essential variability needed for predictive modeling.

Similarly, created a polynomial feature using squared terms for accomodates and minimum_nights in order to capture non-linear relationships. Log transformation was applied to host_total_listings_count to normalize the right skewed distribution and minimize the influence of extreme values. Likewise, ratio features was applied to beds_per_accomodation to analyze how space is utlized as per the accomodation capacity.

Verification of Engineered Features

Summary Statistics of Engineered Features

The MEANS Procedure

Variable	N	N Miss	Mean	Std Dev	Minimum	Maximum
poly_accom	276	0	22.8804348	30.7480704	1.0000000	256.0000000
poly_min	276	0	358.9420290	658.0099228	1.0000000	3600.00
log_host_listings	276	0	1.7458466	1.2176269	0.6931472	7.9330798
beds_per_accom	276	0	1.7605159	0.7003795	0	6.0000000
bath_per_accom	276	0	2.8601881	1.3882583	0	8.0000000

Fig 14: Summary statistics for featured engineered predictors

Frequency of Engineered Features

The FREQ Procedure

bedroom_group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0-1 Bedroom	139	50.36	139	50.36
2-3 Bedroom	113	40.94	252	91.30
4-5 Bedroom	23	8.33	275	99.64
6+ Bedrooms	1	0.36	276	100.00

bathroom_group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0-1 Bath	182	65.94	182	65.94
2-3 Bath	87	31.52	269	97.46
4+ Baths	7	2.54	276	100.00

*Fig 15: Frequency table for newly created categorical va*riables bedroom group and bathroom group

beds_group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0-1 Beds	111	40.22	111	40.22
2 Beds	74	26.81	185	67.03
3-4 Beds	69	25.00	254	92.03
5+ Beds	22	7.97	276	100.00

property_cat	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Entire_Place	207	75.00	207	75.00
Hotel_Commer	9	3.26	216	78.26
Private_Room	53	19.20	269	97.46
Specialty_St	7	2.54	276	100.00

ward_type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Resid	85	30.80	85	30.80
Sub-U	64	23.19	149	53.99
Urban	127	46.01	276	100.00

*Fig 16: Frequency table for newly created categorical va*riables property_category and ward_type

Frequency tables were created for all newly engineered categorical variables to check their distribution and confirm that groupings were balanced. This validation ensured that the transformed features were appropriate for inclusion in the modeling phase.

The data exploration process began with a detailed review of both numeric and categorical predictors. For numeric variables, multicollinearity was assessed using correlation matrices and Variance Inflation Factors (VIF). Categorical variables with many levels were consolidated into fewer, meaningful groups to reduce complexity and improve interpretability. Since the variables

selected for modeling did not contain any missing values, there were no issues related to missing data in the Airbnb analysis.

Multicollinearity Check

Multicollinearity was checked using a Pearson correlation matrix and confirmed through Variance Inflation Factors (VIF). The result obtained demonstrated that polynomial features were highly correlated with their original variables, which is expected. Forward selection was used to choose predictors that added the most value to the model based on their statistical significance and overall contribution to fit. The final model explained about 42% of the variation in price ($R^2 = 0.4211$), with an Adjusted R^2 of 0.4104, and a Root MSE of 0.41794, which indicates a moderate level of predictive accuracy. The VIF of all the varaibles remained close to 1 , which illustrates that selected predictors do not have multicollinearity issues.

Pearson Correlation Coefficients, N = 276
Prob > |r| under H0: Rho=0

	accommodates	minimum_nights	maximum_nights	availability_365	log_host_listings	beds_per_accom	bath_per_accom	poly_accom	poly_min	poly_max	poly_avail
accommodates accommodates	1.00000	-0.23119 0.0001	-0.00394 0.9481	0.00644 0.9151	-0.07571 0.2099	0.23542 <.0001	0.63017 <.0001	0.93440 <.0001	-0.23330 <.0001	0.00317 0.9582	0.03045 0.6145
minimum_nights minimum_nights	-0.23119 0.0001	1.00000	0.03182 0.5987	0.10920 0.0701	0.08076 0.1810	-0.16978 0.0047	-0.22707 0.0001	-0.16510 0.0060	0.93072 <.0001	-0.03822 0.5271	0.09882 0.1014
maximum_nights maximum_nights	-0.00394 0.9481	0.03182 0.5987	1.00000	0.12091 0.0448	0.09890 0.1011	-0.06468 0.2843	0.02029 0.7372	-0.00014 0.9982	0.03305 0.5846	0.96463 <.0001	0.13528 0.0246
availability_365 availability_365	0.00644 0.9151	0.10920 0.0701	0.12091 0.0448	1.00000	0.13427 0.0257	-0.03290 0.5863	-0.03710 0.5394	0.02311 0.7023	0.08423 0.1629	0.06745 0.2641	0.97406 <.0001
log_host_listings	-0.07571 0.2099	0.08076 0.1810	0.09890 0.1011	0.13427 0.0257	1.00000	0.05813 0.3360	-0.02332 0.6997	-0.08415 0.1633	0.07013 0.2455	0.00027 0.9964	0.12536 0.0374
beds_per_accom	0.23542 <.0001	-0.16978 0.0047	-0.06468 0.2843	-0.03290 0.5863	0.05813 0.3360	1.00000	0.27052 <.0001	0.14510 0.0158	-0.13309 0.0271	-0.05592 0.3547	-0.05716 0.3441
bath_per_accom	0.63017 <.0001	-0.22707 0.0001	0.02029 0.7372	-0.03710 0.5394	-0.02332 0.6997	0.27052 <.0001	1.00000	0.48622 <.0001	-0.22245 0.0002	0.01665 0.7830	-0.02022 0.7380
poly_accom	0.93440 <.0001	-0.16510 0.0060	-0.00014 0.9982	0.02311 0.7023	-0.08415 0.1633	0.14510 0.0158	0.48622 <.0001	1.00000	-0.16440 0.0062	0.00743 0.9022	0.05393 0.3721
poly_min	-0.23330 <.0001	0.93072 <.0001	0.03305 0.5846	0.08423 0.1629	0.07013 0.2455	-0.13309 0.0271	-0.22245 0.0002	-0.16440 0.0062	1.00000	-0.02855 0.6368	0.06121 0.3109
poly_max	0.00317 0.9582	-0.03822 0.5271	0.96463 <.0001	0.06745 0.2641	0.00027 0.9964	-0.05592 0.3547	0.01665 0.7830	0.00743 0.9022	-0.02855 0.6368	1.00000	0.07707 0.2018
poly_avail	0.03045 0.6145	0.09882 0.1014	0.13528 0.0246	0.97406 <.0001	0.12536 0.0374	-0.05716 0.3441	-0.02022 0.7380	0.05393 0.3721	0.06121 0.3109	0.07707 0.2018	1.00000

Fig 17: Pearson correlation matrix

Correlation Matrix - Model Variables Only

The REG Procedure
Model: MODEL1
Dependent Variable: log_price

Number of Observations Read	276
Number of Observations Used	276

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	34.30806	6.86161	39.28	<.0001
Error	270	47.16176	0.17467		
Corrected Total	275	81.46982			

Root MSE	0.41794	R-Square	0.4211
Dependent Mean	4.67193	Adj R-Sq	0.4104
Coeff Var	8.94574		

Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	4.17775	0.08880	47.04	<.0001	0
log_host_listings		1	0.03994	0.02087	1.91	0.0567	1.01688
beds_per_accom		1	0.05310	0.03760	1.41	0.1590	1.09156
bath_per_accom		1	0.04924	0.02158	2.28	0.0233	1.41329
poly_accom		1	0.00951	0.00094315	10.09	<.0001	1.32404
poly_min		1	-0.00007680	0.00003958	-1.94	0.0534	1.06774

Fig 18: VIF of numerical variables

Train-Test Split

To evaluate model performance, the dataset was divided into training (80%) and testing (20%) subsets using Simple Random Sampling. A random seed of 42 was applied to ensure reproducibility. The training set contained 221 observations, while the test set included 55 observations, as shown in the verification tables. The FREQ Procedure confirmed that the split was accurate, with each subset representing 100% of its respective selection. We split the data into training and testing sets to check how well the model performs on new, unseen data. The training set is used to build the model, and the testing set is used to evaluate its accuracy. This helps prevent overfitting, where a model works well on the training data but fails on new data.

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling

Input Data Set	LISTINGS_CLEAN
Random Number Seed	42
Sampling Rate	0.8
Sample Size	221
Selection Probability	0.800725
Sampling Weight	0
Output Data Set	FULL_DATA

Fig 19: Summary statement of parameters

Notes (4)

NOTE: There were 276 observations read from the data set WORK.FULL_DATA.

NOTE: The data set WORK.TRAIN has 221 observations and 86 variables.

NOTE: The data set WORK.TEST has 55 observations and 86 variables.

NOTE: DATA statement used (Total process time):

Fig 20: SAS log displaying size of train and test sets

Regression Model

The model was fitted with LASSO (Least Absolute Shrinkage and Selection Operator) regression to identify significant predictors of Airbnb listing prices while addressing multicollinearity issues identified in preliminary analysis. The model was fitted using PROC GLMSELECT with the Schwarz Bayesian Criterion (SBC) as the selection criterion, which automatically determined the optimal number of variables to include. LASSO penalized the least significant variables, automatically setting their coefficients to zero.

Performance measure

For performance measure, Root Mean Squared Error (RMSE) will be used for direct measure of the prediction error in the same units as the response variable. The goal is to minimize the error magnitude and interpret model performance for the real worlkd business settings.

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Value
Model	12	49.94624	4.16219	62.83
Error	208	13.77903	0.06625	
Corrected Total	220	63.72528		

Root MSE	0.25738
Dependent Mean	4.65919
R-Square	0.7838
Adj R-Sq	0.7713
AIC	-364.27820
AICC	-362.23936
SBC	-543.10206

Parameter Estimates		
Parameter	DF	Estimate
Intercept	1	4.528926
property_cat_Private_Room	1	-0.311882
room_type_Entire home/apt	1	0.291460
bedroom_group_0-1 Bedroom	1	-0.199558
bedroom_group_4-5 Bedroom	1	0.338957
bathroom_group_0-1 Bath	1	-0.114474
bathroom_group_4+ Baths	1	0.146890
beds_group_0-1 Beds	1	-0.013558
beds_group_3-4 Beds	1	0.108221
beds_group_5+ Beds	1	0.046159
poly_accom	1	0.001749
poly_min	1	-0.000066471
log_host_listings	1	0.036627

Fig 21: Selected model at step 12

Variables rejected by LASSO:

- Location indicators: ward_type (Residential, Sub-Urban, Urban)
- Host reputation: host_is_superhost, instant_bookable
- Review metrics: number_of_reviews
- Availability metrics: maximum_nights, availability_365
- Polynomial features: poly_max, poly_avail
- Ratio features: beds_per_accom, bath_per_accom

Although feature engineering created four polynomial features, LASSO selected only poly_accom and poly_min, rejecting poly_max and poly_avail as they didn't improve predictions beyond the selected variables. The optimal model was selected at Step 12 (SBC = -543.10), achieving R^2 = 0.784 (explaining 78.4% of price variance) with RMSE = 0.257. The lasso regression model achieved an R-squared of 0.79 by selecting only 12 variables out of 38 potential variables. The optimal model was selected at step 12 where all statistical criteria like AIC, AICC and SBC and adjusted R-squared were converged. The model obtained the RMSE of 0.253 which implies that

the average error in the price listing is $23.62 per night, which demonstrates the model strong predictive accuracy.

Twelve predictors selected at the final model:

Predictor Variable	Coefficient	Effect on Price
Intercept	4.529	Baseline: $92.60/night
property_cat_Private_Room	-0.312	-27%
room_type_Entire home/apt	+0.291	+34%
bedroom_group_0-1 Bedroom	-0.200	-18%
bedroom_group_4-5 Bedroom	+0.339	+40%
bathroom_group_0-1 Bath	-0.114	-11%
bathroom_group_4+ Baths	+0.147	+16%
beds_group_0-1 Beds	-0.014	-1%
beds_group_3-4 Beds	+0.108	+11%
beds_group_5+ Beds	+0.046	+5%
log_host_listings	+0.037	+4% per doubling
poly_accom	+0.001749	Shrinks towards 0
poly_min	-0.000066	Shrinks towards 0

Variable selection process

The coefficient progression graphs demonstrate the crucial insights as revealed by the trajectory lines and optimal stopping points marked as selected steps where the SBC reaches the lowest –543.10. The graph shows that the Airbnb price is highly dominated by the room_type_Entire home/apt (+0.39) meaning entire home charges approximately 40% more, property_cat_private_room dropped to –0.31 meaning it charges 27% less. Likewise, 4-5 bedroom_group charges around 40% and studios charge 18% less as shown by the dropping orange line. Similarly, variables like ward_type, review_counts, instant_booking are concentrated near zero lines which implies these variables don't have significant impact on pricing. From which we can say a 4-bedroom house in a residential area charges the same as an identical house downtown. Therefore, well-configured properties with more bedrooms and bathrooms impacts pricing more than the location.

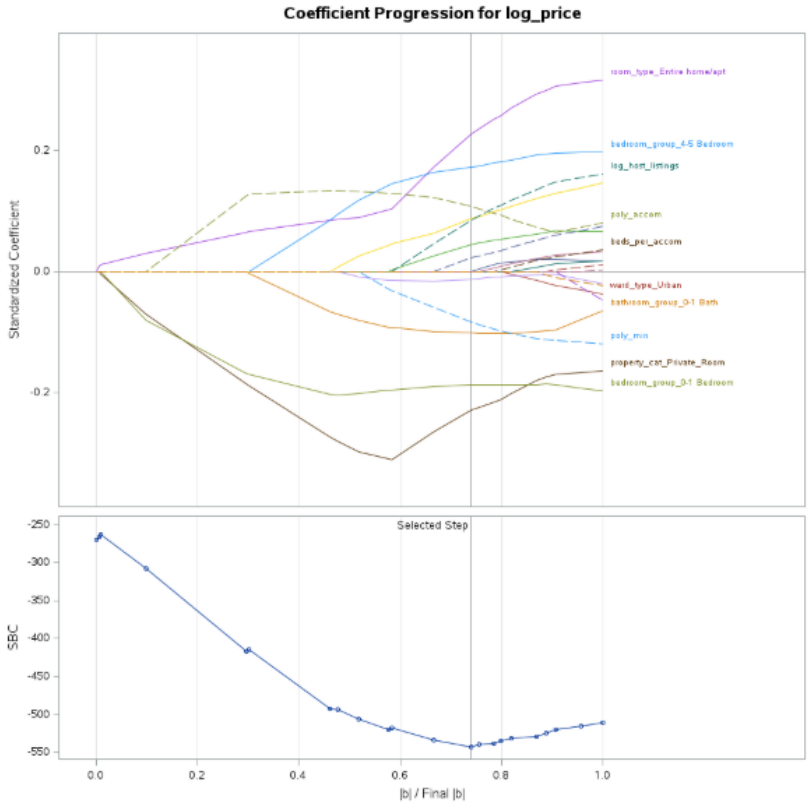

Fig 22: Coefficient progression for log_price

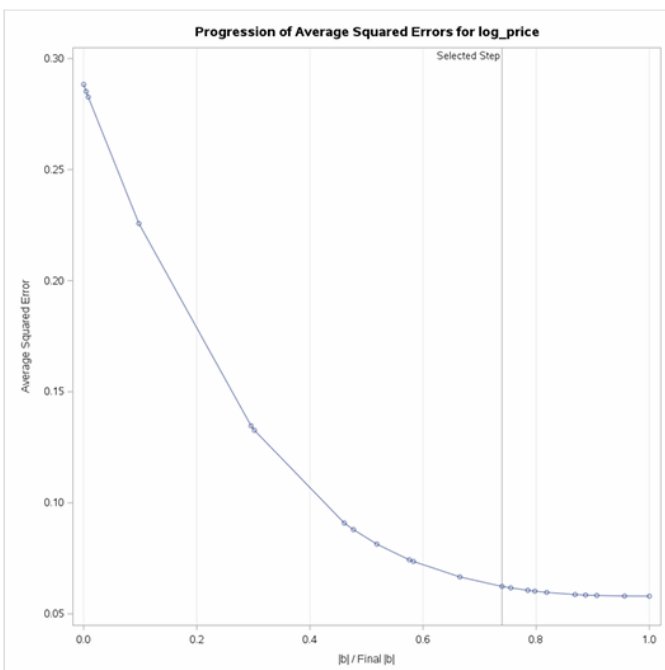

Fig 21: Progression of average squared errors log_price

The progression of average squared errors gives us an improvement in prediction accuracy as we add more variables in the model. From the graph, it is evident that the line drops from 0.29 to 0.10 in just 5 steps. The steep decline explains that these variables are most of the variation in price. Likewise, at step 6 the curve flattens reaches 0.06 at step 12, which showcases minor increment in the accuracy from variables like bed_type and host_listings count. After step 12, the graph becomes flat which indicates adding more predictors doesn't significantly improve the model and reduce prediction accuracy.

Performance comparison of training and test accuracy

LASSO - Training RMSE

Obs	_TYPE_	_FREQ_	squared_error	trans_error	true_error	RMSE	trans_RMSE	true_RMSE
1	0	221	13.7790	227.878	1222.95	0.24970	1.01544	5.53370

Fig 23: RMSE of LASSO in Train data

LASSO - Test RMSE

Obs	_TYPE_	_FREQ_	squared_error	trans_error	true_error	RMSE	trans_RMSE	true_RMSE
1	0	55	6.20062	62.6966	1124.37	0.33577	1.06768	20.4431

Fig 24: RMSE of LASSO in Test data

LASSO Model - R-Squared Training Data

The REG Procedure
Model: MODEL1
Dependent Variable: log_price

Number of Observations Read	221
Number of Observations Used	221

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	50.25976	50.25976	817.41	<.0001
Error	219	13.46552	0.06149		
Corrected Total	220	63.72528			

Root MSE	0.24796	R-Square	0.7887
Dependent Mean	4.65919	Adj R-Sq	0.7877
Coeff Var	5.32205		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	-0.39954	0.17772	-2.25	0.0256
p_log_price	1	1.06575	0.03798	28.59	<.0001

Fig 25: RMSE of LASSO in Train data

Metric	Training Set	Test Set
R-squared	0.78	0.69
RMSE (log scale)	0.24	0.33
RMSE(dollar scale)	$23.62 per night	$30 per night

Table 1: Performance comparison in Train and Test Data

The performance of training and test case showcase that R-squared in Test set decreases by 3% from 0.78 to 0.69. The RMSE increases from $23.62 to $30 per night, which is increased by $6. After analyzing training and test sets it is clear the model is not generalized for training and test sets, and it doesn't overfit the model.

Residual diagnostics for Linear Regression

To evaluate the reliability of the LASSO model, residual diagnostics is performed to check whether the model's behavior is unbiased and generalizable for implementing the real-world setting.

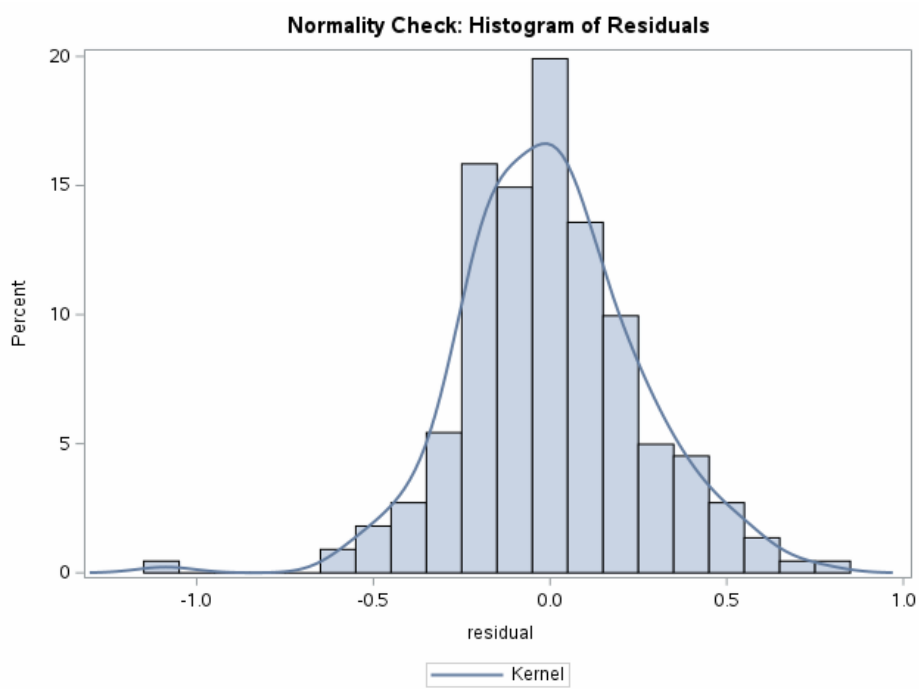

Fig 26: Histogram of the residuals

The histogram illustrates that the residuals are noramlly distributed. Most residuals are seen clustered around 0 which shows models have no bias. On avearage we can say that the prediction from the model is unbiased.

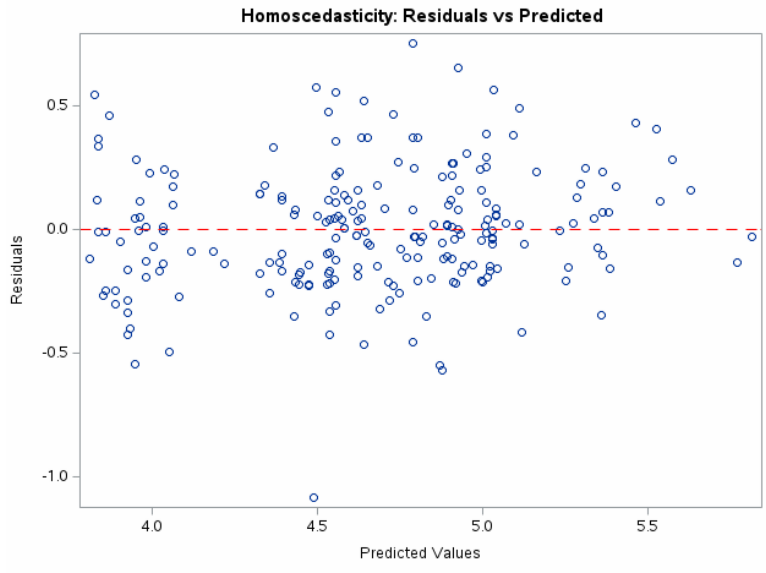

Fig 27: Histogram of the residuals

The residuals vs the predicted plot displays the constant variance 0.063 across all predicted values. The validates the assumption of constant error variance with standard errors; confidence intervals and hypothesis tests are reliable. Therefore, we can conclude that the LASSO model pricing recommendations.

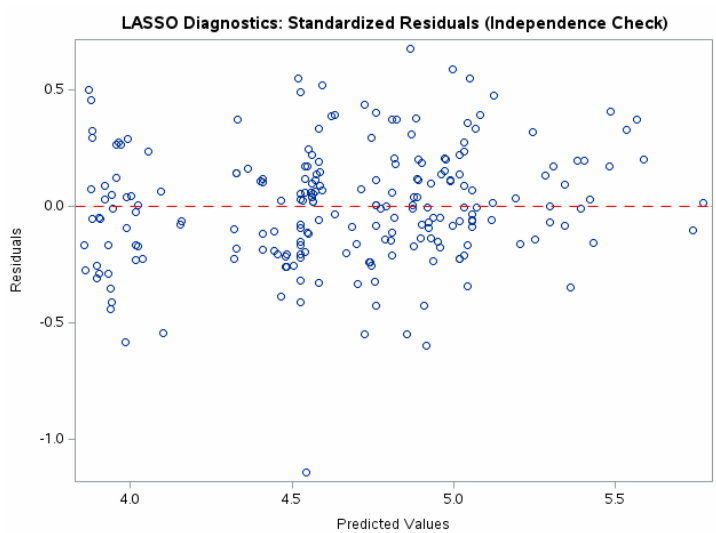

Fig 28: Standardize residuals independence check

The standardized residuals plot reveals that the out of 216 out of 221 observations falls under the acceptable standard deviation range, with only around 5% of the outliers which indicates the continuous opportunities for the improvement with few cases requiring the human invention in price prediction.

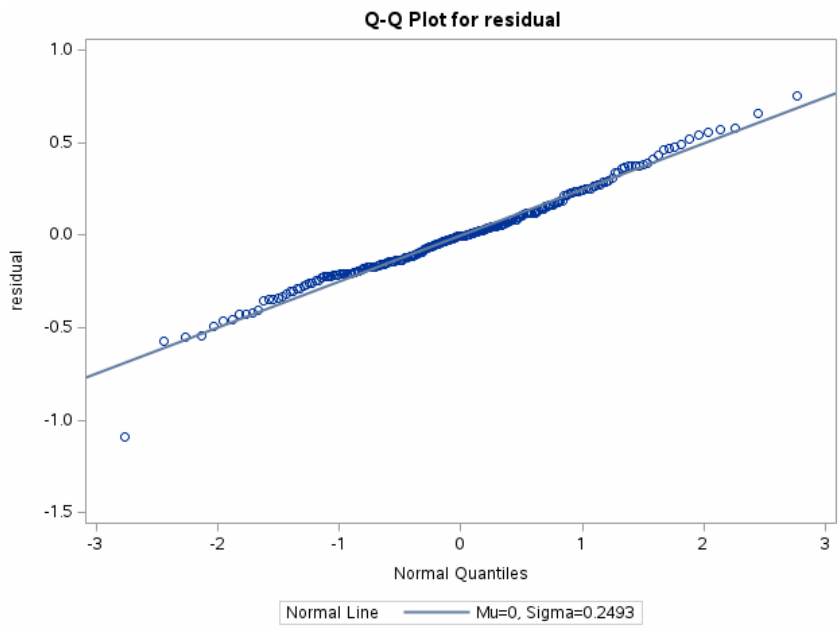

Fig 29: Q-Q plot for residuals

The Q-Q plot is one of the diagnostic tools that was used to check if the model follows the normal distribution so that we can evaluate if we can rely on our confidence interval, p-values, and risk calculations. The diagonal line is the representation of how the error is supposed to be distributed, and the blue dots are the actual error distribution. From the Q-Q plot, we can say that the majority of the points are close to the diagonal line, implying that the errors are normally distributed. This also validates that the log transformation successfully normalized the right-skewed price distribution.

Decision Trees

The HPSPLIT Procedure

Performance Information	
Execution Mode	Single-Machine
Number of Threads	2

Data Access Information			
Data	Engine	Role	Path
WORK.TRAIN	V9	Input	On Client
WORK.TREE_TRAIN_SCORED	V9	Output	On Client

Model Information	
Split Criterion Used	Variance
Pruning Method	Cost-Complexity
Subtree Evaluation Criterion	Cost-Complexity
Number of Branches	2
Maximum Tree Depth Requested	10
Maximum Tree Depth Achieved	10
Tree Depth	4
Number of Leaves Before Pruning	142
Number of Leaves After Pruning	7

Number of Observations Read	221
Number of Observations Used	221

Fig 30: Summary of HPSPLIT Decison tree

For the decision, tress HPSPLIT was used for the tree splitting. The model used variance as the split criterion to minimize variability in predicted prices and used cost-complexity pruning to control overfitting. The maximum tree depth achieved is 10 and it contains 142 leaves which demonstrate complex structure. After pruning, the tree was simplified to a depth of 4 with only 7 leaves, making it more interpretable and less prone to overfitting.

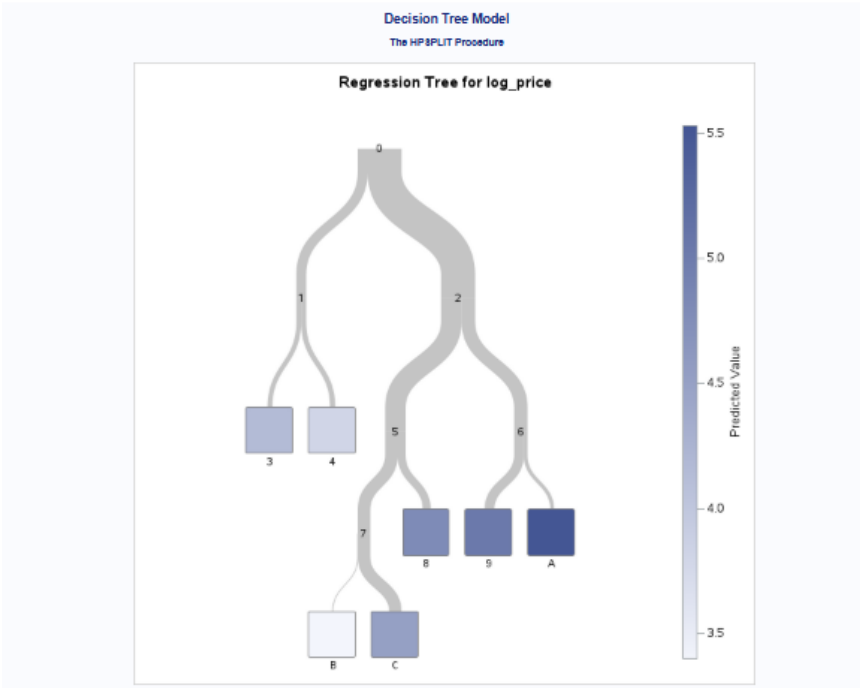

Fig 31: Regression tree for log_price

This figure shows the decision tree built for predicting the log_price. The split denotes the decision based on feature values and the branches lead to nodes with predicted price range where darker shades represent the higher price prediction around 5.5, and the lighter shades represent the lower predicted price around 3.5.

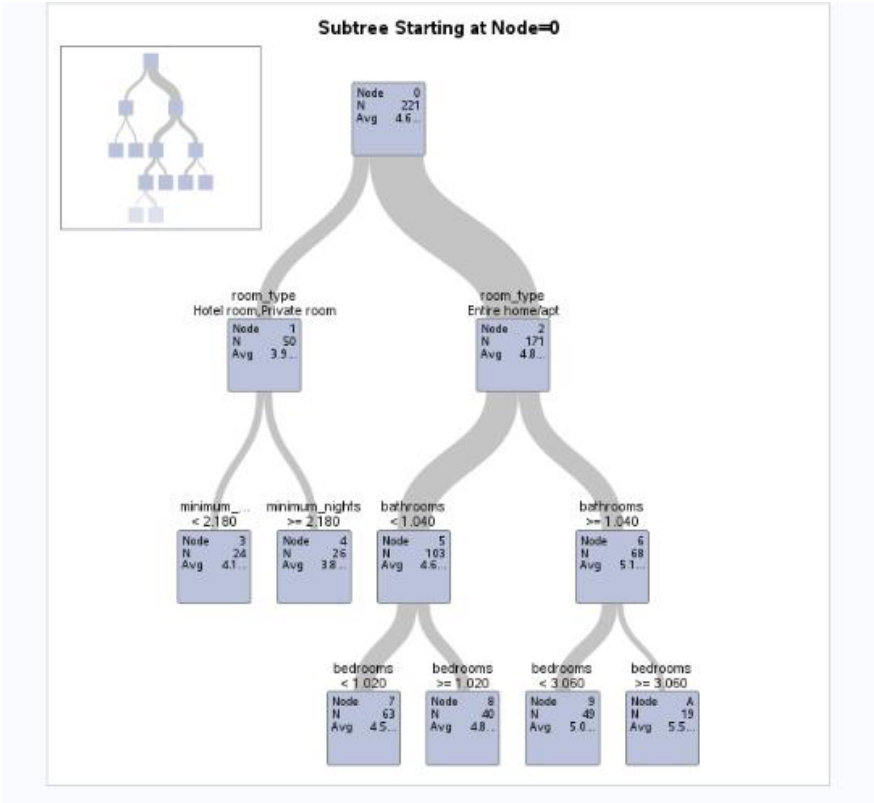

Fig 32: Subtree starting at node=0

The figure shows the subtree created using the HPSPLIT, the root node contains all the 221 records with an average log_price of 4.6. The first split occurs on **room_type**, dividing "Hotel room/Private room" from "Entire home/apt. Similarly, further split occurs on minimum_nights, bathrooms and bedrooms. For example, Airbnb listings which have more than three bathrooms and bedrooms have the highest predicted prices and listings with less nights, bedrooms and bathrooms have significantly lower prices.

Evaluation metrics

Decision Tree Model

The HPSPLIT Procedure

Model-Based Fit Statistics for Selected Tree		
N Leaves	ASE	RSS
7	0.0649	14.3437

Variable Importance				
		Training		
Variable	Variable Label	Relative	Importance	Count
room_type	room_type	1.0000	5.3652	1
bathrooms	bathrooms	0.6914	3.7094	2
bedrooms	bedrooms	0.4275	2.2938	2
minimum_nights	minimum_nights	0.2339	1.2550	1

Fig 33: Subtree starting at node=0

The variable importance highlights the room_type as the most influential predictors followed by bathrooms, bedrooms, and minimum_nights. These preditors plays an important role while determining the price where room_type has the highest relative importance of 1.00.

Random Forest

In order to implement Random Forest in our analysis, HPFOREST was used which allowed the flexible process to customize the model with the use of hyperparameter values. The Vars to try indicates the number of variables to consider splitting in the node which is 7, maxtrees represents the number of trees in the forest, which is 500, Maxdepth represents the depth of a node in a tree which is 20, leafsize represents the smallest number of training observations which is 6, Alpha represents threshold p-value for the significance level of test of association of a candidate varaible with target which is 0.1.

Model Information		
Parameter	Value	
Variables to Try	7	
Maximum Trees	500	
Actual Trees	500	
Inbag Fraction	0.6	(Default)
Prune Fraction	0	(Default)
Prune Threshold	0.1	(Default)
Leaf Fraction	0.00001	(Default)
Leaf Size Setting	6	
Leaf Size Used	6	
Category Bins	30	(Default)
Interval Bins	100	
Minimum Category Size	5	(Default)
Node Size	100000	(Default)
Maximum Depth	20	
Alpha	0.1	
Exhaustive	5000	(Default)
Rows of Sequence to Skip	5	(Default)
Split Criterion	.	Variance
Preselection Method	.	BinnedSearch
Missing Value Handling	.	Valid value

Variable Importance Random Forest

Loss Reduction Variable Importance					
Variable	Number of Rules	MSE	OOB MSE	Absolute Error	OOB Absolute Error
bedrooms	787	0.049119	0.04765	0.055699	0.051303
room_type	306	0.046657	0.04493	0.044971	0.043786
property_cat	254	0.044026	0.04391	0.042267	0.041664
accommodates	560	0.037805	0.03116	0.036609	0.028345
bathrooms	500	0.021019	0.02067	0.023861	0.022274
beds	484	0.015119	0.01226	0.015770	0.011615
minimum_nights	1090	0.006226	0.00076	0.011264	0.003176
instant_bookable	119	0.000394	-0.00034	0.000434	-0.000631
host_is_superhost	245	0.000671	-0.00076	0.000979	-0.001016
host_total_listings_count	695	0.003331	-0.00086	0.005311	-0.001128
ward_type	344	0.001391	-0.00102	0.001929	-0.001384
maximum_nights	406	0.001351	-0.00142	0.001989	-0.001876
number_of_reviews	1186	0.003987	-0.00340	0.006017	-0.003689
availability_365	1123	0.003807	-0.00420	0.006098	-0.005460

Fig 34: Loss reduction variable importance

The table illustrates the variable importance for the Random Forest Model on loss reduction. The importance is measured using metrics like Mean Squared error and Out of Bag (OOB) error, which represents how much the variables contribute to reducing the prediction error. From the loss reduction variable importance table, we can say that the bedroom is the most important in predicting the price with OOB MSE of 0.047 followed by the room_type and property_cat.

Hyperparameter Tuning

The hyperparameter for both Decision Tree and Random Forest were selected to balance model complexity and predictive accuracy. A maximum depth of 10 was allowed to capture the

complexity and cost-complexity to decrease the overfitting and increase the model interpretability. Likewise, split criterion was set to variance for regression tasks since it minimizes the variability within the nodes. The Vars_to_try option was kept to 7 to add variety to the trees, minimizing correlation and enhancing generalization. Alpha was set at 0.1 as a significance level for variable selection during splitting.

Residuals diagnostics for Tree model

Model	Metric	Value
Decision Tree	Average Squared Error (ASE)	0.0649
Decision Tree	Residual Sum of Squares (RSS)	14.34
Random Forest	Out-of-Bag (OOB) Error	0.072
Random Forest	Training RMSE	0.2189
Random Forest	Test RMSE	0.3071

Table 2: Residuals diagnostics for Tree model

To test and compare the models, the study used measures that measure prediction accuracy. The decision tree's Average Squared Error (ASE) was 0.0649, corresponding to an RMSE of about 0.2548, and the Residual Sum of Squares (RSS) was 14.34, indicating the total error over all observations.

After 500 trees, the Random Forest's Out-of-Bag (OOB) error stabilized at roughly 0.072, which is consistent with the test RMSE of 0.3071, indicating good generalization. The train RMSE for Random Forest was 0.2189, which was somewhat lower than that of the Decision Tree, indicating that it had better prediction performance. While ASE and RSS are beneficial for tree complexity, RMSE provides a more interpretable metric in the same units as the target variable,

Model Performance Comparison

Below is the model performance comparison for all three models from which it is evident that Random Forest is the best performing model with RMSE of 0.30 in the test data set. Likewise, the gap between the training and test data set is around 0.09, which is not significantly large, so overfitting is limited.

Model	Training_RMSE	Test_RMSE
LASSO	0.2497	0.3358
Decision Tree	0.2548	0.3096
Random Forest	0.2189	0.3071

Table 3: Model performance comparision table

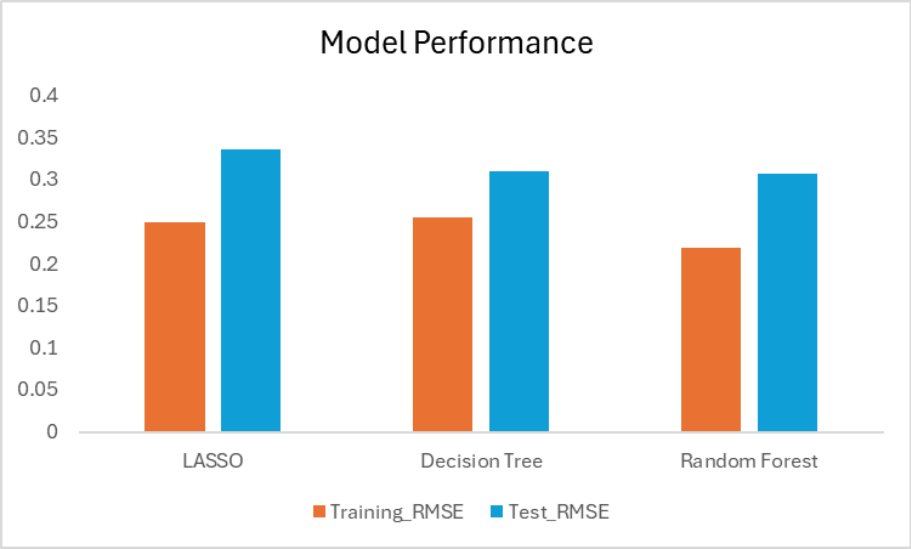

Fig 35: Model Performance of LASSO, Decision tree and Random Forest

Future Improvements and Business Implications

For enhancing the pricing accuracy and delivering actionable insights for Airbnb hosts and property managers, several improvements can be incorporated in future. To reduce the risks of overfitting and ensure the robustness of the model, cross validation can be applied. Likewise, hyperparameter tuning for decision trees and random forests can help improve predictive accuracy. Seasonality and demand trends will also help in dynamic pricing strategies, which will help in adjusting pricing strategies for high travel times, and ensure maximum revenue. Likewise, one of the limitations of this analysis was the small dataset, which contains only 351 observations, which can reduce the model's ability to generalize and explain patterns in pricing behavior.

Appendix

Overfitting analysis

Overfitting Analysis

Model	Train_RMSE	Test_RMSE	RMSE_Difference	Percent_Increase	Overfitting_Level
Random Forest	0.2189	0.3071	0.0882	40.29	High Overfitting
Decision Tree	0.2548	0.3096	0.0548	21.51	Moderate Overfitting
LASSO	0.2497	0.3358	0.0861	34.47	High Overfitting

Fit Statistics of Random Forest showing 40 trees.

Fit Statistics			
Number of Trees	Number of Leaves	Average Square Error (Train)	Average Square Error (OOB)
1	16	0.07441	0.11099
2	33	0.06061	0.09501
3	50	0.05434	0.08383
4	67	0.05222	0.07511
5	82	0.05090	0.07655
6	99	0.04774	0.07577
7	117	0.04818	0.07853
8	135	0.04840	0.07958
9	152	0.04857	0.07885
10	169	0.04925	0.07792
11	186	0.04958	0.08125
12	203	0.04897	0.07831
13	220	0.04902	0.07694
14	236	0.04930	0.07574
15	253	0.04928	0.07664
16	269	0.04939	0.07528
17	284	0.04975	0.07551
18	300	0.04973	0.07605
19	317	0.04976	0.07530
20	333	0.05018	0.07609
21	349	0.04980	0.07563
22	365	0.05010	0.07595
23	381	0.04979	0.07496
24	399	0.04972	0.07547
25	416	0.04976	0.07582
26	433	0.04977	0.07566
27	449	0.04977	0.07538
28	466	0.04995	0.07516
29	483	0.05012	0.07488
30	500	0.04994	0.07474
31	516	0.04977	0.07461
32	534	0.04985	0.07497
33	552	0.04965	0.07448
34	569	0.04965	0.07458
35	587	0.04958	0.07504
36	604	0.04946	0.07523
37	622	0.04963	0.07569
38	639	0.04951	0.07564
39	656	0.04947	0.07557
40	674	0.04967	0.07577